

Claude Code App Prompt

Interactive Seating Chart & Classroom Manager

System Role & Objective

You are an expert full-stack developer. Your task is to build a modern, interactive, and fully responsive Seating Chart web application designed for a Smartboard and laptop.

Tech Stack Requirements

- **Frontend:** React (Next.js or Vite), Tailwind CSS for styling, and Framer Motion for smooth animations.
- **Backend/Database:** Supabase or Firebase. *Crucial:* The app must have real-time synchronization so changes made on a laptop instantly reflect on the Smartboard. Set up the schema and prompt me for API keys to initialize this.
- **Icons:** Lucide React or similar modern icon library.
- **Drag & Drop:** `@dnd-kit/core` or similar library that supports both mouse and touch sensors.

Core UI & Layout Constraints

- The application must be a Single Page Application (SPA) that fits exactly to the full screen (100vh, 100vw).
- **No Scrolling:** Use CSS Grid and Flexbox to ensure the layout dynamically resizes to fit any screen without ever requiring vertical or horizontal scrolling.
- **Tactile UX:** All buttons must have hover, active (press), and focus states with subtle scale animations to feel tactile.
- **Auto-Save:** Any state changes (seating, timers, class creation) must automatically sync and save to the database. Provide an intuitive dropdown or tab system at the top of the UI to seamlessly switch between multiple classes.

Feature 1: The Desks & Classroom Grid

- Create a grid of 30 desks (5 columns wide x 6 rows deep).
- **Desk UI:** Each desk must display the assigned Student's Name (large, highly legible font) and Homeroom Number.
- **Color Coding:** Dynamically style desks based on the student's gender. Boys = soft light blue (`#ADD8E6` or Tailwind `blue-200`), Girls = soft light pink (`#FFB6C1` or Tailwind `pink-200`). Empty desks should be a neutral gray.

Feature 2: Class Settings & Roster Management (Modal)

Include a "Class Settings" button that opens a clean modal. Capabilities inside the modal:

- **Name/Rename Class:** Text input to update the current class name.
- **Roster Upload/Input:** Provide a drag-and-drop zone for a CSV file (Columns: Student Name, Homeroom Number, Gender) to bulk import students. Also include a manual entry form to add students one by one.
- **Edit Students:** A list of the current roster where clicking a student allows editing their Name, Homeroom, or Gender.
- **Create New Class:** Button to generate a new class and empty seating chart without deleting existing ones.
- **Delete Class:** Button to delete the current class. Must require a strict fail-safe (e.g., typing the class name to confirm or a double-confirmation modal).
- **Unseat All:** Button that removes all students from their assigned desks in the current class, placing them into an unassigned "Roster Pool" (a sidebar or bottom drawer waiting to be seated).

Feature 3: Seating & Swapping

- Implement a "Swap Seats" toggle button.
- **Tap-to-Swap:** When active, tapping a desk selects it (trigger a pulsing animation or glowing border). Tapping a second desk swaps the two students. The mode stays active until the toggle button is pressed again.
- **Drag-and-Drop:** Additionally, allow dragging a student from one desk (or the unassigned roster) and dropping them onto another desk. This must work flawlessly for both mouse (laptop) and touch (Smartboard).

Feature 4: Random Pickers

- **State Management:** The app must track `pickedStudents` and `pickedRows` per active session to ensure no repeats until everyone/every row has been picked.
- **Pick a Student Button:** When clicked, trigger a "Jeopardy-style" rapid flashing animation across all unpicked desks for 4-5 seconds. End by highlighting a single random desk brightly. Tap the highlighted desk to dismiss/deselect it.
- **Pick a Row Button:** Teachers call columns "rows". Group the columns (1 through 5) visually. Animate the columns flashing randomly for 4-5 seconds, finally highlighting one entire column. Tap the column to dismiss.

Feature 5: The Flip-Card Timer

- Build a countdown timer styled exactly like a mechanical flip-clock (minutes and seconds).
- **Constraints:** Max minutes = 99. Max seconds = 60.

- **Controls:** A hidden slide-down menu underneath the clock containing `+` and `-` buttons for setting minutes and seconds, along with Start, Pause, and Stop buttons.
- **Settings Gear Icon:** Opens a timer settings modal:
 - **25% Warning Toggle:** When toggled on, the flip-clock numbers/background must turn red when the timer reaches the final 25% of its original set time.
 - **Alarm Sounds:** Dropdown to select an alarm sound: Ding, Chime, Bell, Train Whistle, Guitar, Rooster.
 - *Important Audio Instruction:* Do not ask me for audio files. Use the browser's native `Web Audio API` (Oscillators, GainNodes, etc.) to synthetically generate these sounds, or pull free, reliable public CDN audio links.

Final Instruction for Claude Code:

Please scaffold the project, set up the file structure, and begin implementing the code step-by-step. Let me know when you need me to set up the database project to provide the environment variables.